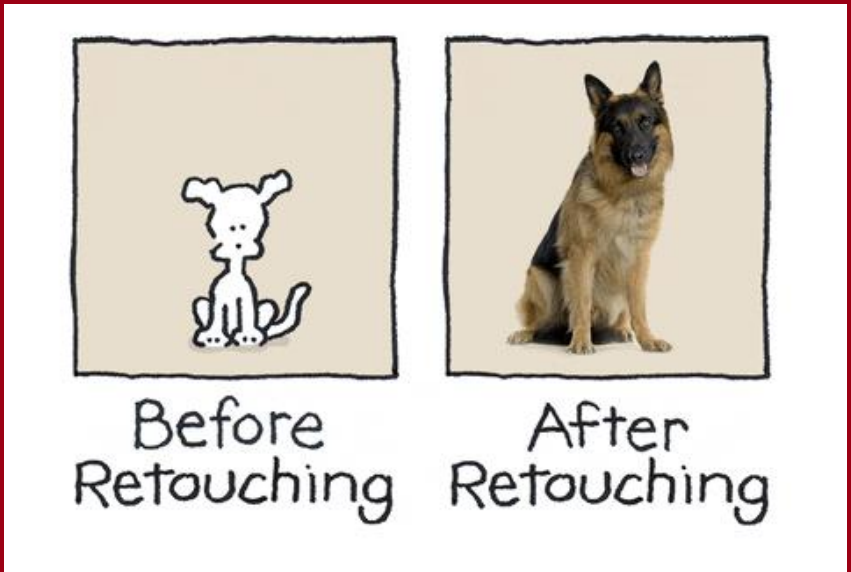
Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

A.A. 2026/2027

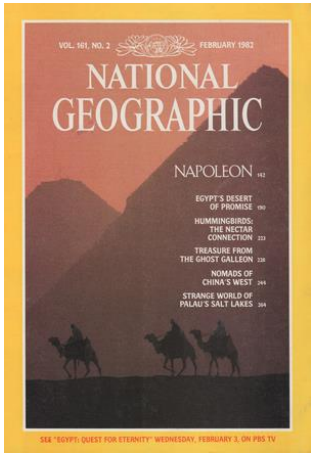
Lecture 12 – Multimedia integrity

Simone Milani
Room 216
DEI A

Phone: 049 827 7641
E-mail:
simone.milani@dei.unipd.it

DIGITAL EDITING OF PICTURES: PICTURES AND INFORMATION

1982: First example of digitally-altered picture



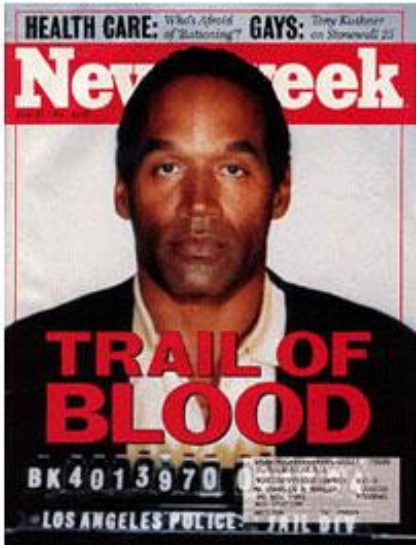


2003

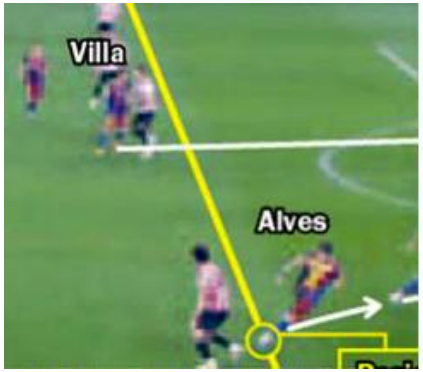


1994



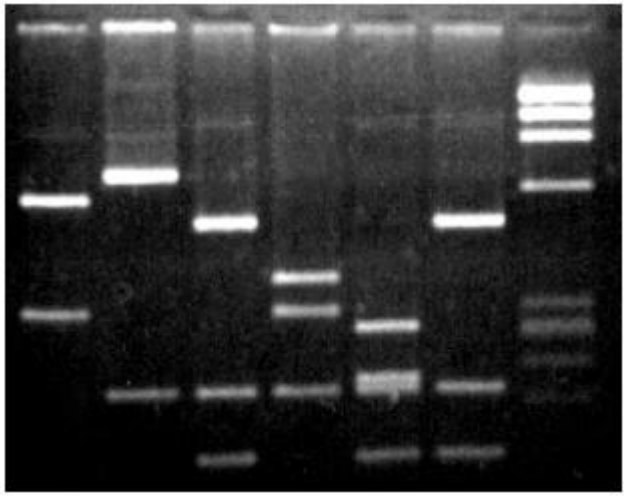


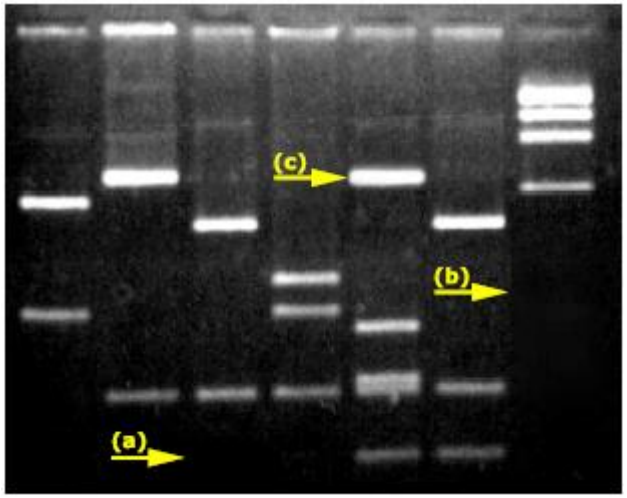
2011

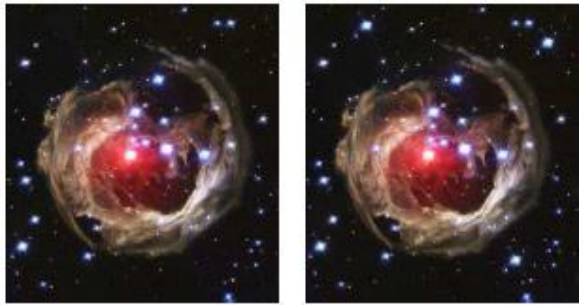


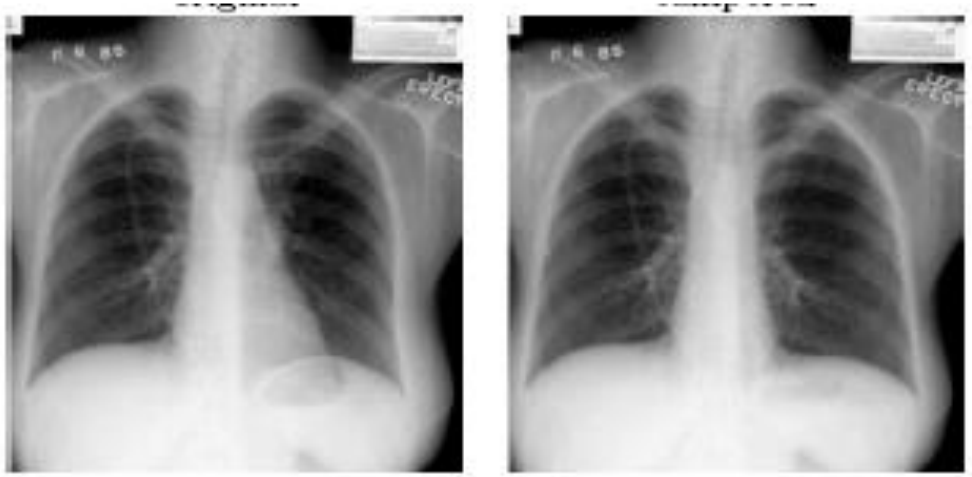


SCIENTISTS DO FAKES TOO…









FAKE IMAGES FROM COMPUTER GRAPHICS

CG



CG



Real



CG



Real

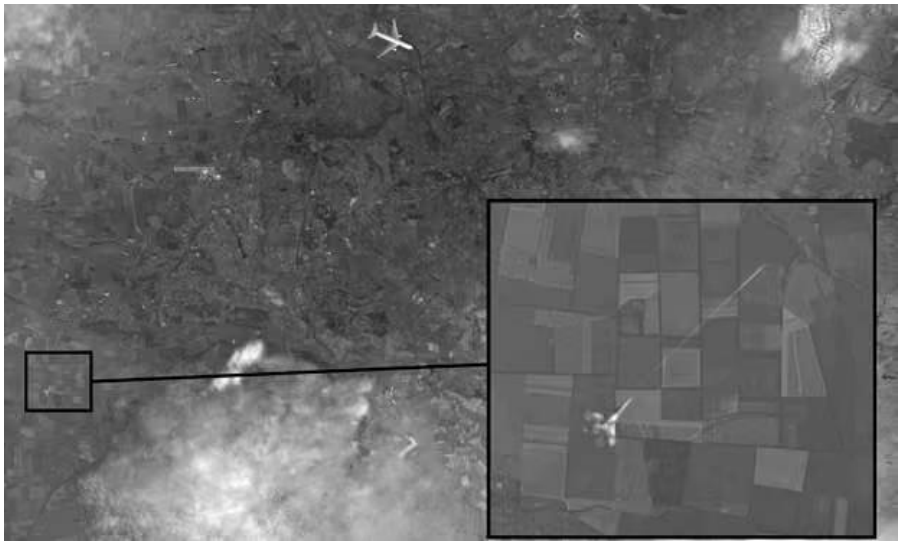




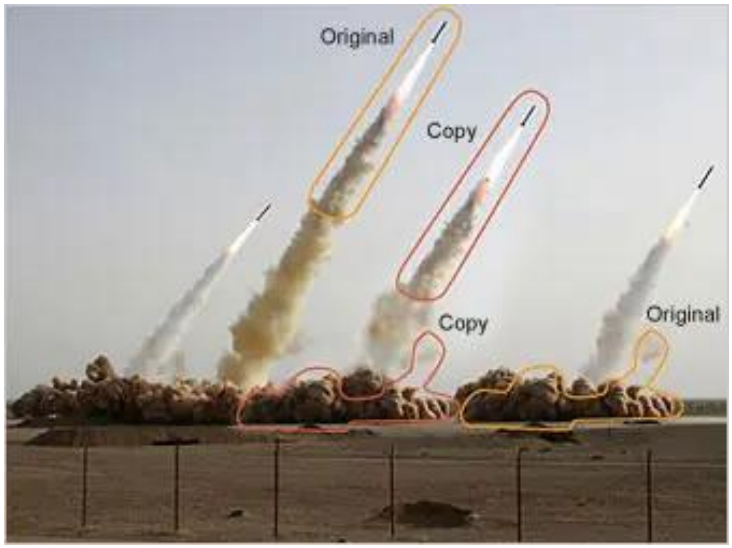
IMAGE FORGERY

Manipulation: changing of any image's objects or contents without giving any indication that anything has changed.

One of the most popular techniques = **image splicing:** *adding or hiding things by copying portions of one image and pasting them elsewhere in the same image or into a different image.*



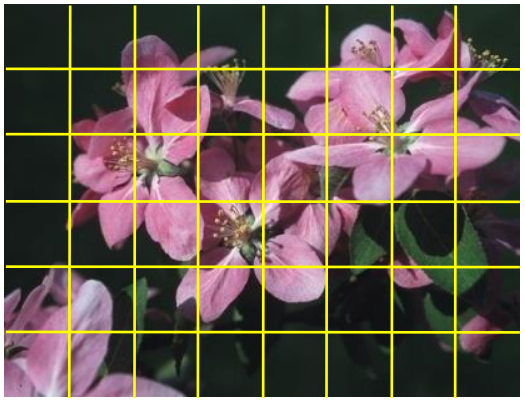
2014: Malaysia Airlines disaster

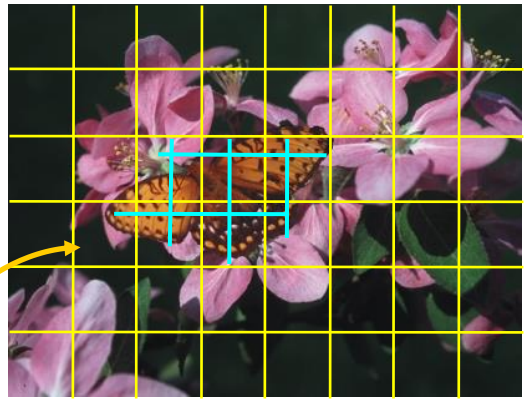


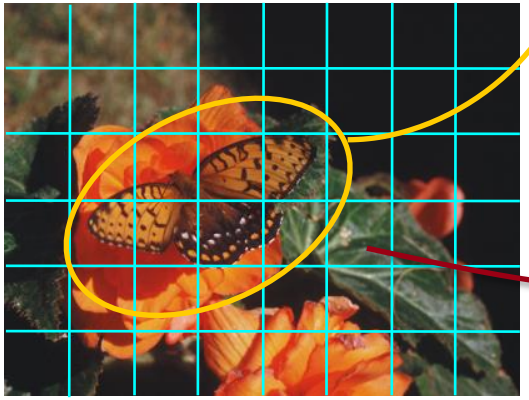
2012: Nuclear Tests

BASIC PRINCIPLE: HOMOGENEITY

Most of the tools described before permits detecting alterations on images: the fact is that the outputs of these detectors are not the same on all the regions of the image.







$$PRNU_1 \neq PRNU_2$$
$$CFA_1 \neq CFA_2$$
$$DJPEG_1 \neq DJPEG_2$$

CHECK CONSISTENCY/INCONSISTENCY OF FOOTPRINTS

- PRNU
- CFA pattern
- Interpolation characteristics
- Local correlation
- Coding parameters
- Double/multiple vs. single compression
- Editing traces
- Physical phenomena (shadows, motion, illumination, ….)
- Geometric inconsistencies
- Similarity/dissimilarity between different parts
- Similarity dissimilarity between different images

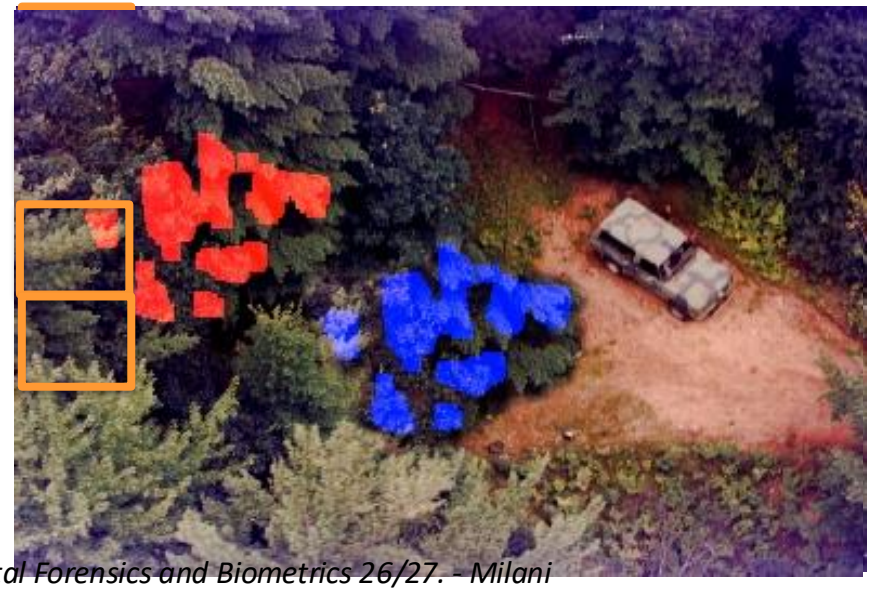
Whenever elements expected to be similar are different, raise alarm!

SIMILARITY ALARM: DUPLICATE ELEMENTS

- Sometimes similarity is a problem: a part of the image is copied and replaced
- Useful to hide something







How to check?

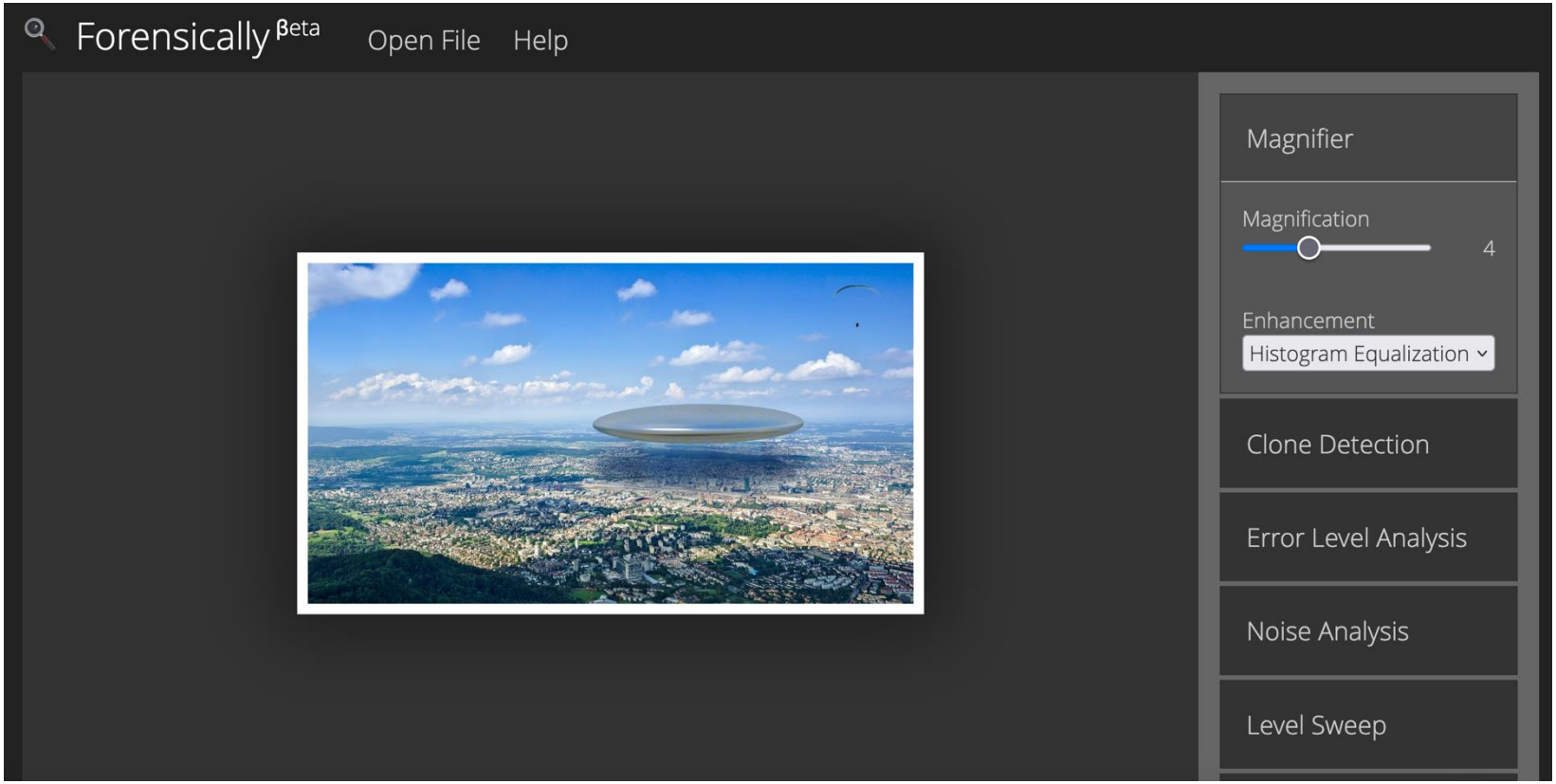
- Scan the image with windows

(or SIFT descriptor to find similar points)

- When points are the same: alarm
- No way to find which is the original part

FORENSICALLY EXAMPLE

https://29a.ch/photo-forensics/#forensic-magnifier



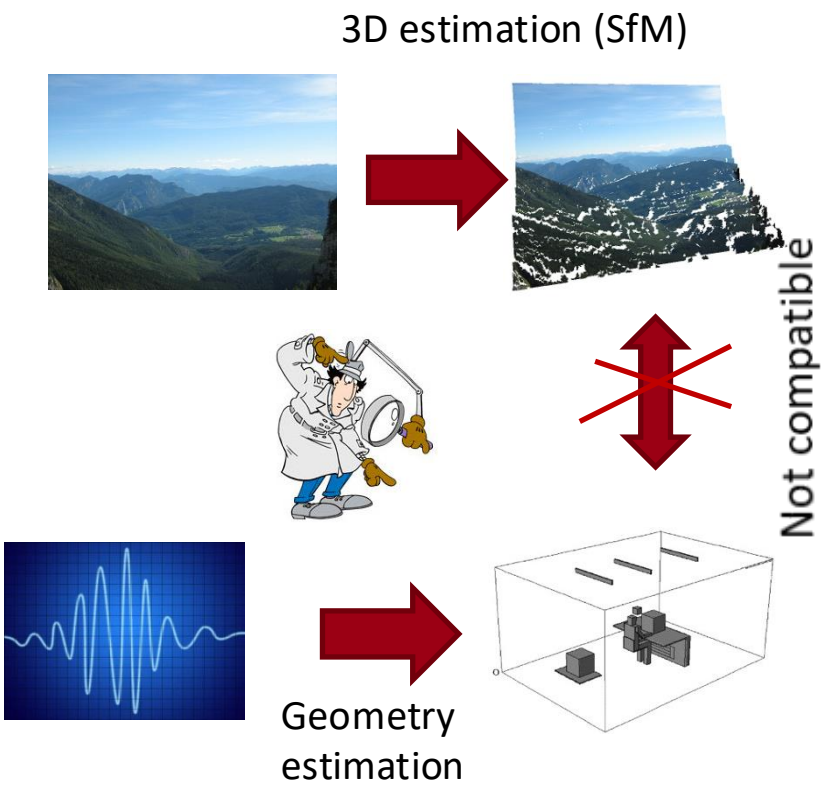
CONSISTENCY FROM PHYSICS AND REAL ENVIRONMENTS

A different sets of strategy, checks the physical characteristics of the scene

- Light and illumination
- Shadows
- Perspective
- Reflections
- Motion (for video)



ENVIRONMENT ESTIMATION

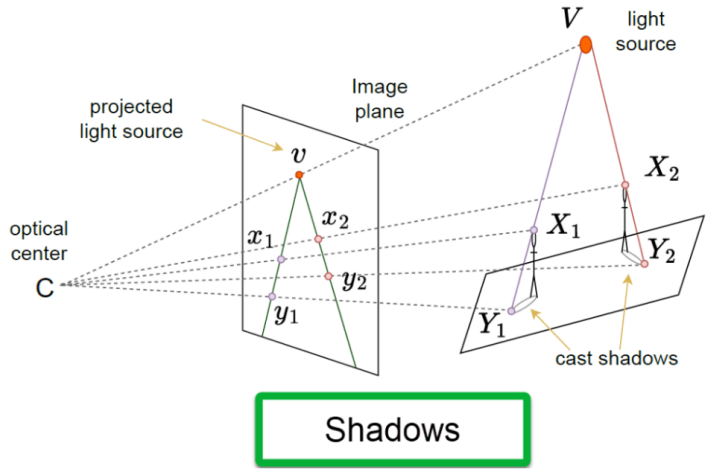


It is possible to estimate the geometry of the environment from audio and video.

In case it is not consistent, a fake is detected.



SHADOWS AND PERSPECTIVE



Shadows

Parallel lines obtained by connecting corresponding points in a reflection are projected in lines intersecting in the corresponding vanishing point onto the 2D image.





Original



Fake

INCIDENT LIGHT





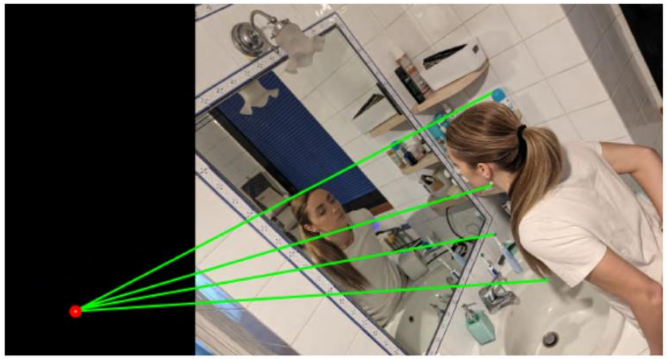
- It is possible to mode the direction of the incoming light (under some assumption)

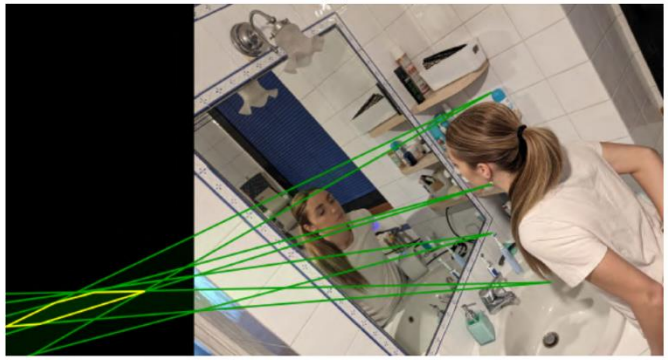
- Including an object in a picture keeping the illumation consistent is quite difficult
- Automatic check!





REFLECTIONS

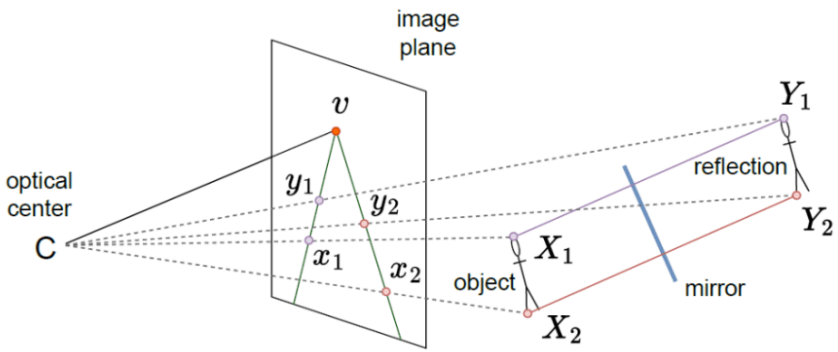




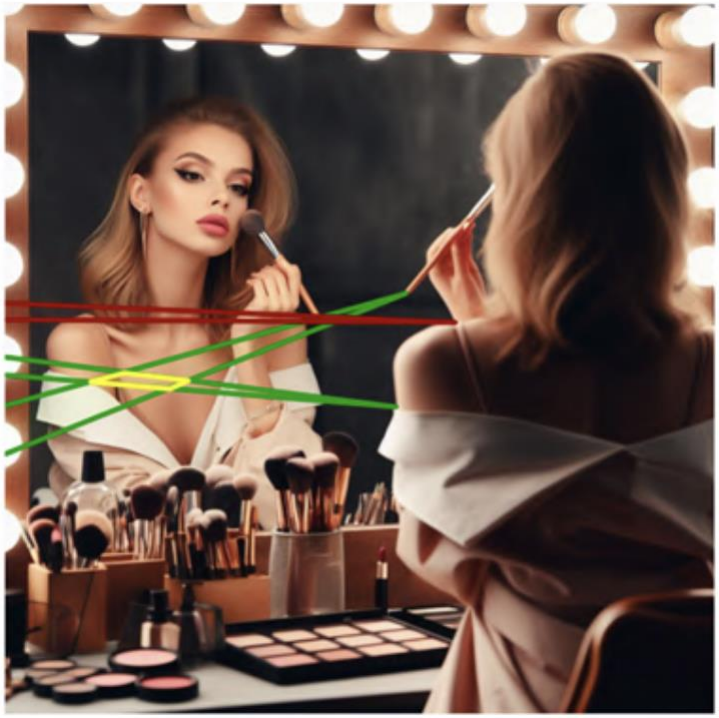
In authentic images, rays intersect in the vanishing point corresponding to the direction of the projected mirror normal

The intersection of relaxed constraints: compatible region for the reflection vanishing point.

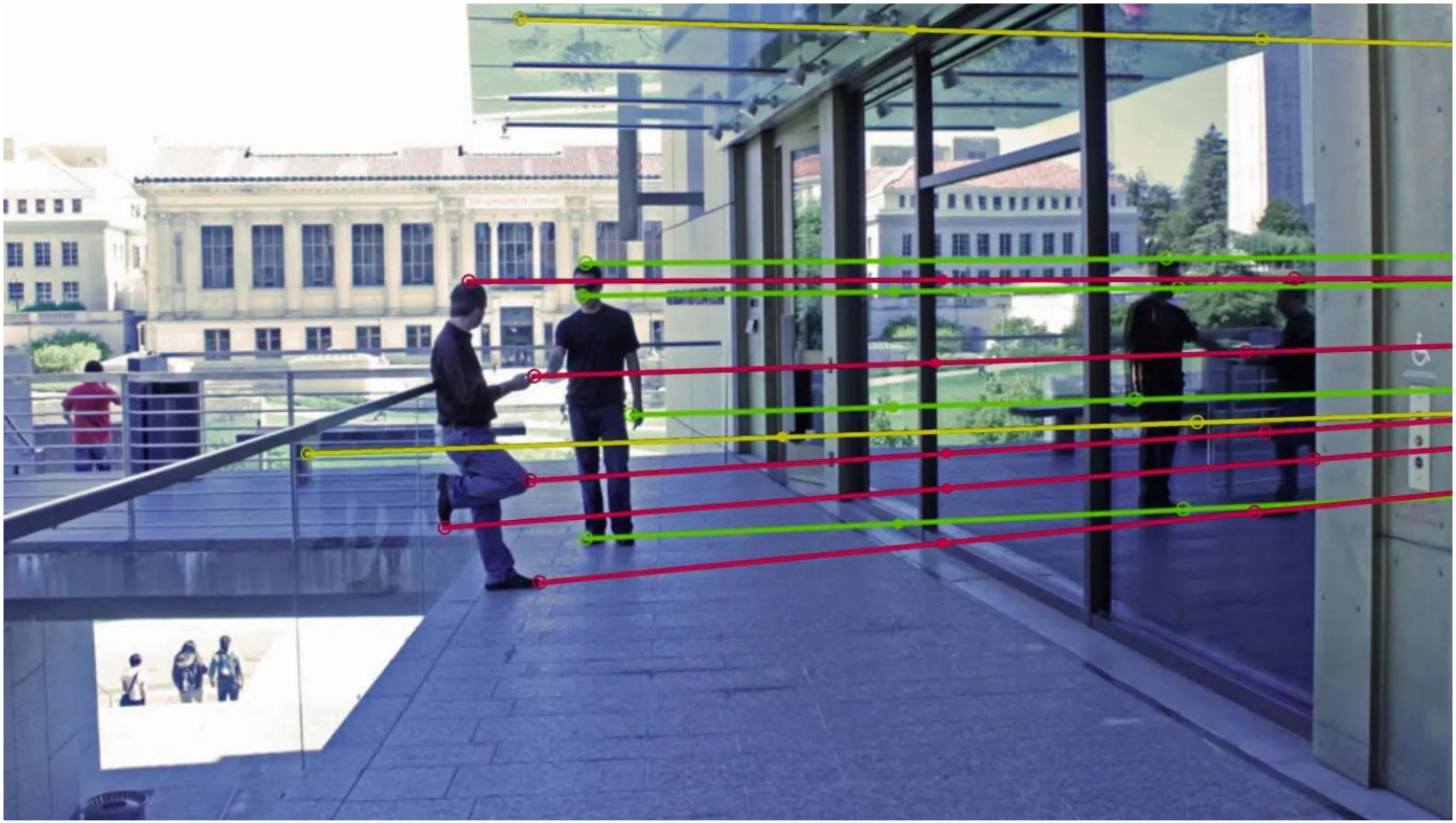
For synthetic image, intersection is not granted



Reflections



EXAMPLE



VIDEO EDITING



VIDEO EDITING



VIDEO EDITING

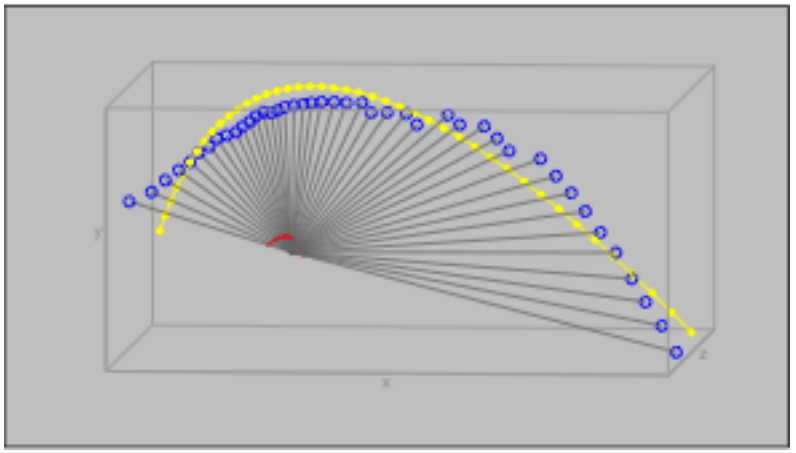


BALISTIC MOTION



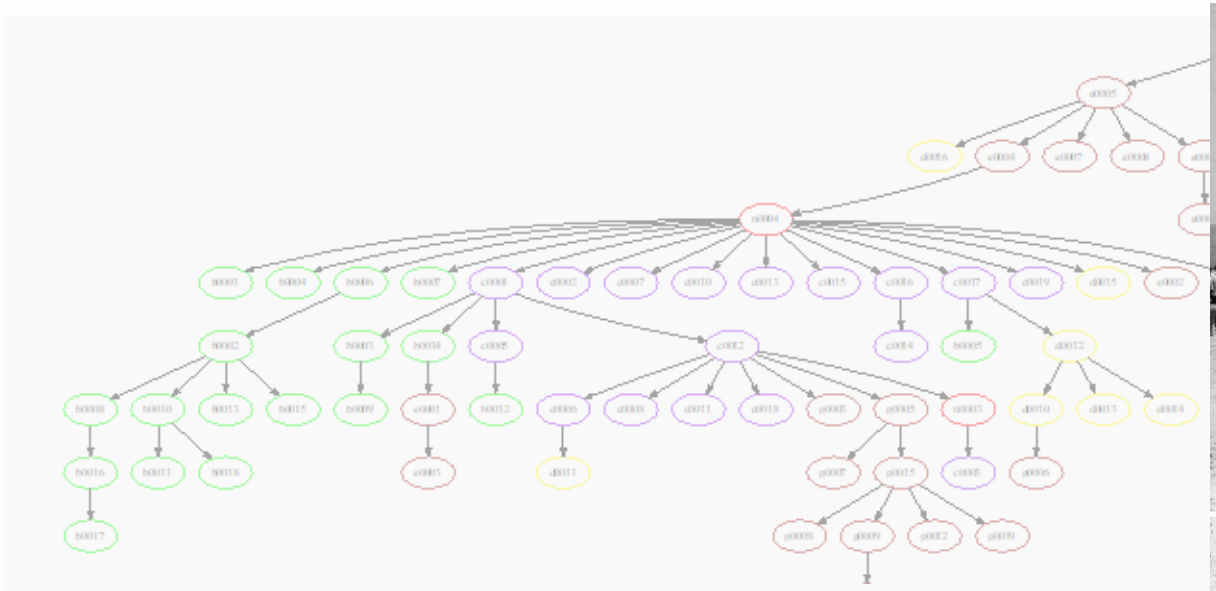
BALISTIC MOTION

- After compensating the perspective distortion, it is possible to build a parabolic motion model (yellow)
- Tracking the object on video and comparing the trajectories highlight the fake!



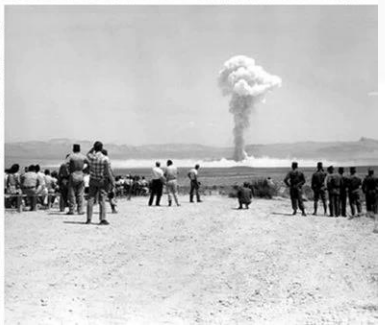


MULTIPLE-ORIGIN TRACKING















MULTIMEDIA FORGERY AS A CYBERSECURITY ISSUE

- Forged media contents are nowadays very diffused

- Deep learning has dramatically increased the phenomenon

  Legal use
  - Video editing and retouching in movie industry
  - Generation of new products, contents, characters
  - Artistic and entertainment uses
  - Assisted design

Manual editing Deepfake



  Illegal use
  - Use of deepfakes in propaganda, public opinion manipulation
  - Creation of evenge porn material, illegal content production
  - Generation of fake audio files mimicking other people voice for authentication

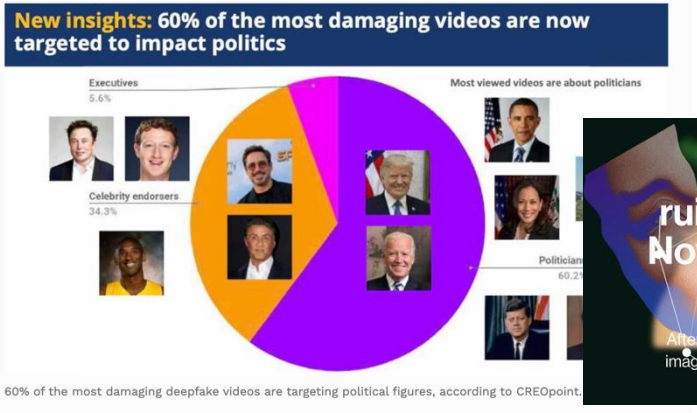

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PRO CYBER NEWS

Fraudsters Used AI to Mimic CEO's Voice in Unusual Cybercrime Case

Scams using artificial intelligence are a new challenge for companies





SUMMARY OF DEEPFAKE TOOLS

Tools	Link	Features
Faceswap	https://github.com/deepfakes/faceswap	Using two encoder-decoder pairs. Parameters of the encoder are shared.
Faceswap-GAN	https://github.com/shaoanlu/faceswap-GAN	Adversarial loss and perceptual loss (VGGface) for auto-encoder
Few-Shot Face Translation	https://github.com/shaoanlu/fewshot-face-translation-GAN	use a pre-trained face recognition for GAN processing. Incorporate semantic priors from FUNIT and SPADE
DeepFaceLab	https://github.com/iperov/DeepFaceLab	Expand from the Faceswap method with new models Support multiple face extraction modes
Dfaker, DeepFake_tf	https://github.com/dfaker/df https://github.com/StromWine/DeepFake tf	DSSIM loss function Keras and TF implementation
AvatarMe	https://github.com/lattas/AvatarMe	Reconstruct 3D faces from "in-the-wild" images. 4K by 6K 3D faces from a single low-resolution image
DiscoFaceGAN	https://github.com/microsoft/DiscoFaceGAN	Generate face images of virtual people with independent latent variables of identity, expression, pose, and illumination.
StyleRig	https://gvv.mpi-inf.mpg.de/projects/StyleRig	Portrait images with a rig-like control over a pretrained and fixed StyleGAN. (self-supervised without manual annotations).
FaceShifter	https://lingzhili.com/FaceShifterPage	Face swapping in high-fidelity
FSGAN	https://github.com/YuvalNirkin/fsgan	Face swapping and reenactment model
"Do as I Do" Motion Trans-fer	https://github.com/carolineec/EverybodyDanceNow	Automatically transfer the motion from a source to a target person by learning a video-to-video translation
Neural Voice Puppetry	https://justusthies.github.io/posts/neural-voice-puppetry	Audio-driven facial video synthesis

GAN DETECTION

- Artifacts on hair/eyes: high frequency details





- Face geometry artifacts
  - Smooth cheecks and forehead
  - Eyes/eyebrows, shadows
  - Glasses: is there a glare?
  - Facial hair/mole
  - Blinking
  - Lips' color

Note that some of these detections can be overcome with a proper training [e.g., blinking example]. link



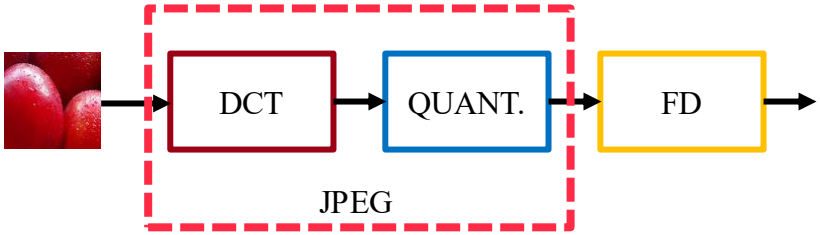


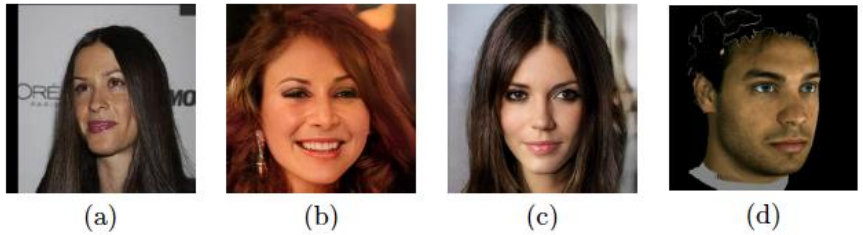
STATISTICAL ANOMALY

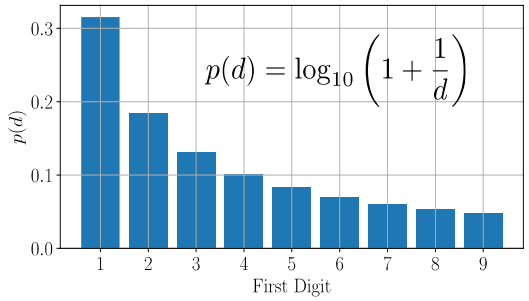
- GAN images may have different statistics from natural images
- Benford's Law can capture these traces

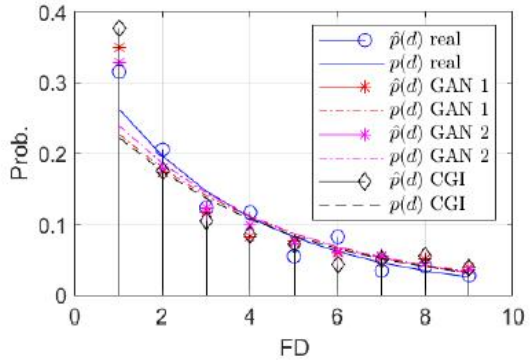
$$d_{b,n,\Delta}(k) = \left\lfloor \frac{|c_{n,\Delta}(k)|}{b^{\lfloor \log_b |c_{n,\Delta}(k)| \rfloor}} \right\rfloor$$

- Given a natural image (JPEG compressed), compute the first digit (FD) of quantized DCT coefficients. It is known that distribution of the FDs follows Benford's law.









SUMMARY OF DEEPFAKE DETECTORS

Tools	Techniques	Features
Eye blinking	LRCN	Model blnking pattern (temporal).
Intra-frame and temporal inconsistencies	CNN/LSTM/RCN/ResNet/VGG/Capsule Networks	Analyze the temporal pattern (more generally); Define general rules or learn from sample data
Face warping artifacts	VGG16, ResNet50	resolution inconsistency between the warped face area and the surrounding context
MesoNet	CNN	Analyze videos at mesoscopic level.
Capsule forensics	Capsule Network	Multiple capsule network to grasp temporal artifacts
Head Poses	SVM	Extract 68 dlib landmarks; use classifier to check if fake or not
Eye, teach and facial features	Logistic regression and Neural Networks	Look for details around mouth and eyes.
PRNU	Use camera/image forensic features	Check for inconsistencies between features on camera-generated and GAN-generated images.
Phoneme-viseme mismatches	CNN	Mismatch between mouth dynamics and phonemes
Face X-Ray	CNN	Find boundaries between real face and fake image.
Use fake data dectors	Benford's law	Check if statistical anomalies are present in image data

DATA HIDING:
WATERMARKING AND STEGANALYSIS

INTUITIVE EXAMPLE

"My friend Bob: Until yesterday I was using binoculars for stargazing. Today I decided to try my new telescope. The galaxies in Leo and Ursa Major were unbelievable! Next, I plan to check out some nebulas and then prepare to take a few snapshots of the new comet. Although I am satisfied with the telescope, I think I need to purchase light pollution filters to block the xenon lights from a nearby highway to improve the quality of my pictures. Cheers, Alice."

① Take the initials

mfbuyiwubfstidttmnttgilaumwuniptcosnatpttafsotncaiaswttitintplpftbtxlfanhtitqompca

② Filter with π=0.3141592653689793...

buubdlupnpsspx

③ Take the previous letter

ATTACK TOMORROW

WATERMARK IN HISTORY

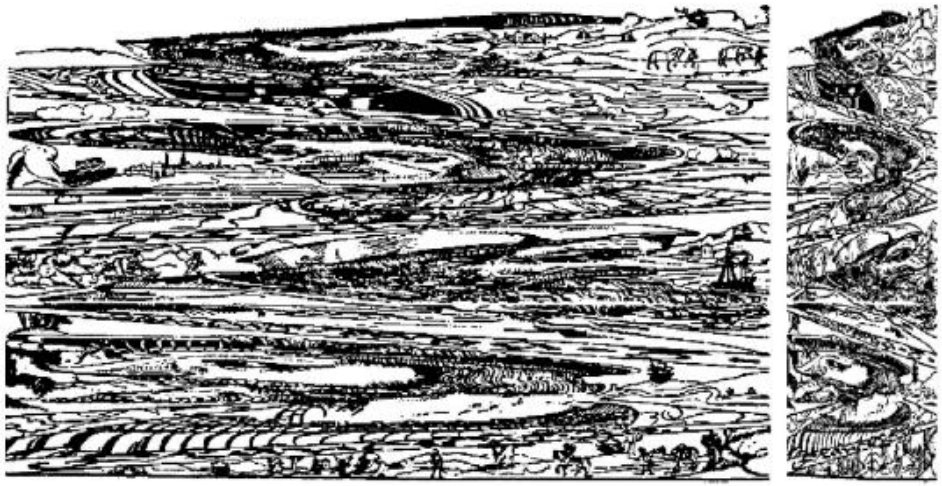


The Ambassadors
Hans Holbein
16th century

WATERMARK IN HISTORY



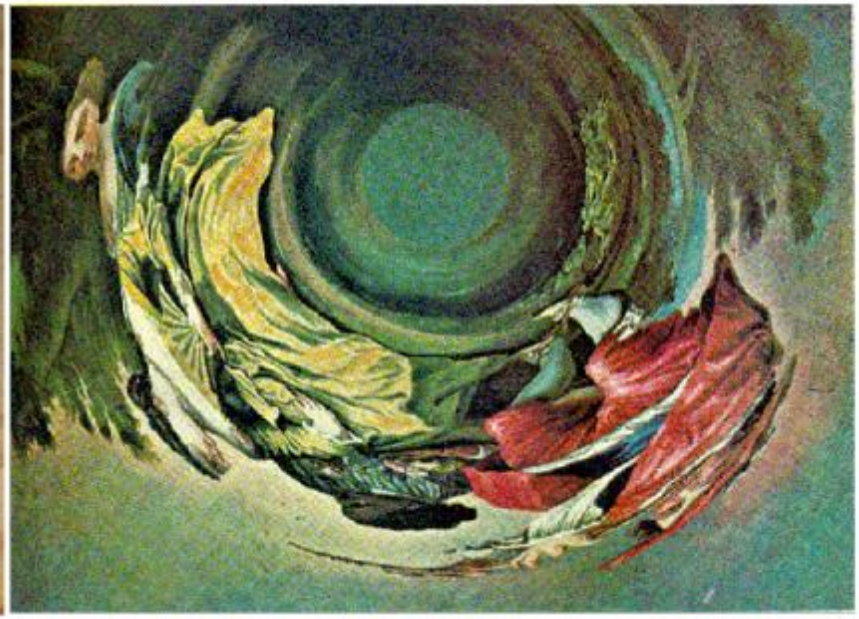
WATERMARK IN HISTORY



Vexierbild
Schoen
16th century
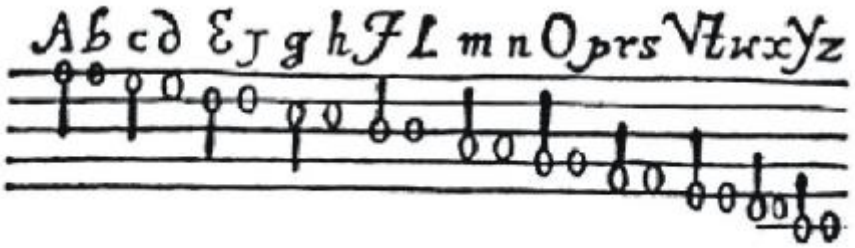
Cylindrical Anamorphism
18th century





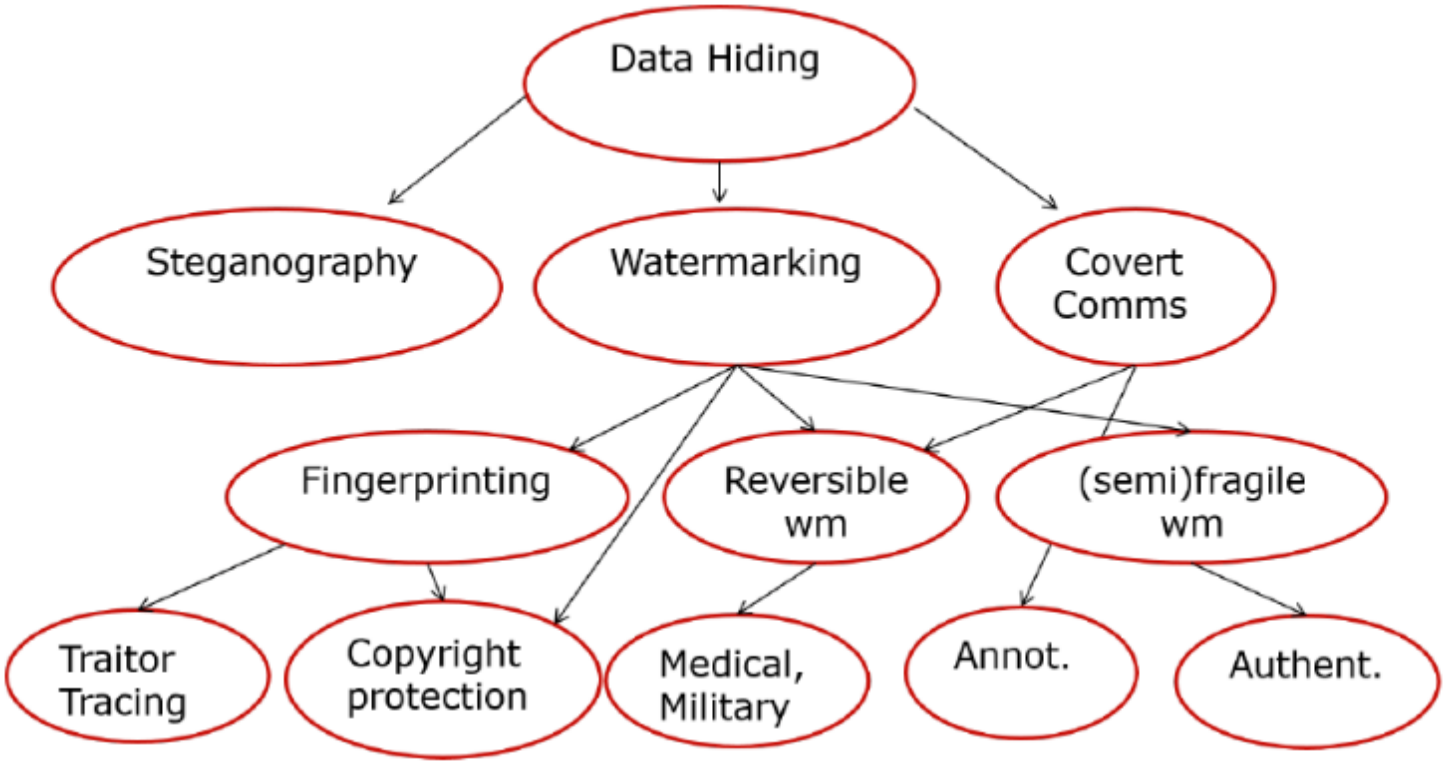
WATERMARK IN HISTORY

- Linguistic Steganography: Boccaccio's 'Amorosa Visione'.
- 'Hypnerotomachia Poliphili' (1499).
- John Wilkins's method (1694) -> Bach.



- Cardan's grid.
- First system for 'copyright protection': Claude Gellée de Lorraine (ca. 1635).

A TAXONOMY OF WATERMARKING



SOME TERMS ...

- **Watermark:** It is a message that is adhered to an object, which may carry complementary information and does not alter the object semantics.
- **Visible/Invisible watermarks.**
- **Imperceptible data hiding:** Harder to eliminate; does not affect the quality of the object to be protected.
- **Steganography:** Hiding a message into an object. Low detectability. Robustness is not important.
- **Attack:** Operation that aims to eliminate, distort or extract the hidden data in an unauthorized way. Can be intentional or non-intentional.

WHY NOT CRYPTOGRAPHY?

- Cryptography solves the problem of secure distribution, but it is not enough.
- Managing access rights: conditional access systems.
- Once access to digital media is granted, cryptography does not prevent from making copies.
- Integrity protection and illegal use identification: watermarking.
- Both technologies are complementary, not competing.

Instead watermarking …

- ✓ Typically, **blind**: recovery without the original.
- ✓ **Robust**: Resilient to both intentional and non-intentional attacks.
- ✓ **Imperceptible**: Must take into account human senses.
- ✓ **Security and keys**: Access must be forbidden if there is no access to the keys; key generation, distribution and management.
- ✓ **Conflict resolving**: Must stand multiple watermarking.
- ✓ **Payload**.

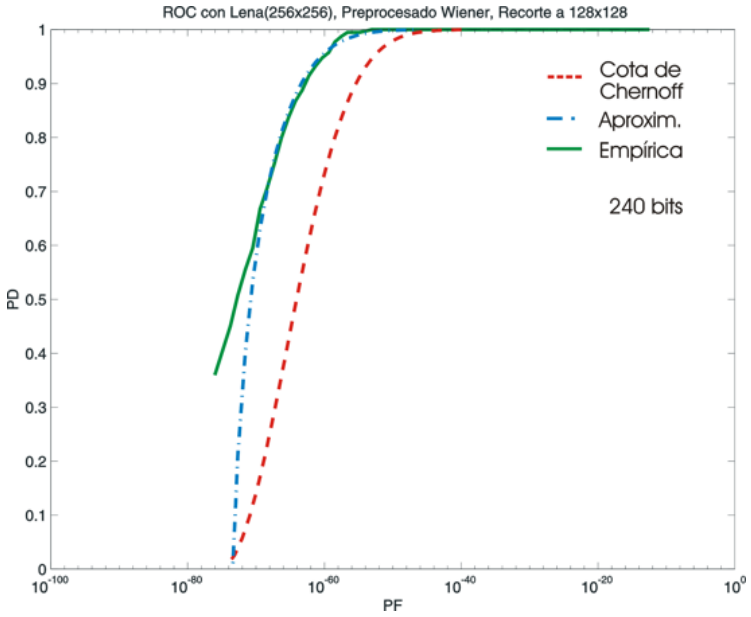
WATERMARKING VS. DATA HIDING

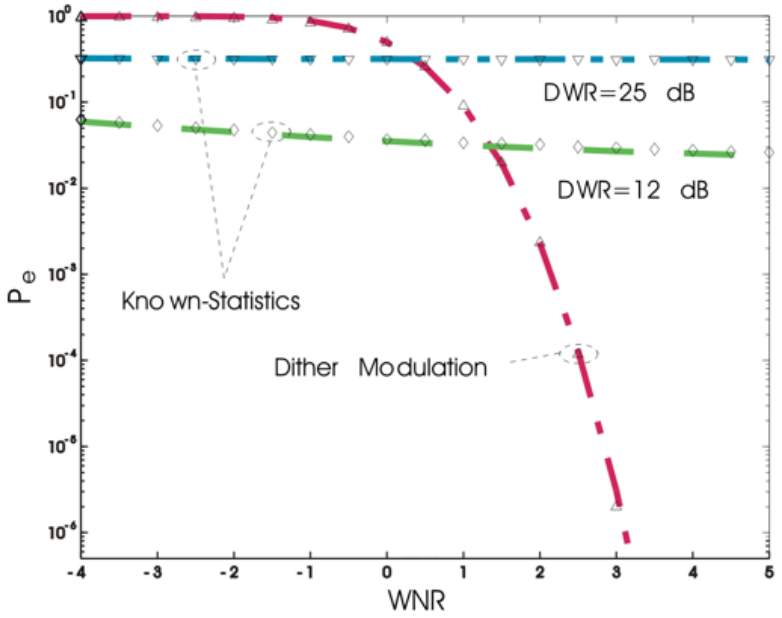
Watermarking:

- Embed a key-dependent watermark (maybe unique) and detect its presence.
- Binary hypothesis test (yes/no answer).
- Performance: ROC.

Data hiding:

- Embed multiple bits of information in a key-dependent way and decode them.
- Multiple hypothesis test.
- Performance: BER





TWO FAMILIES OF WATERMARKING

Additive spread-spectrum (Cox, 1996): add the watermark to the host signal.

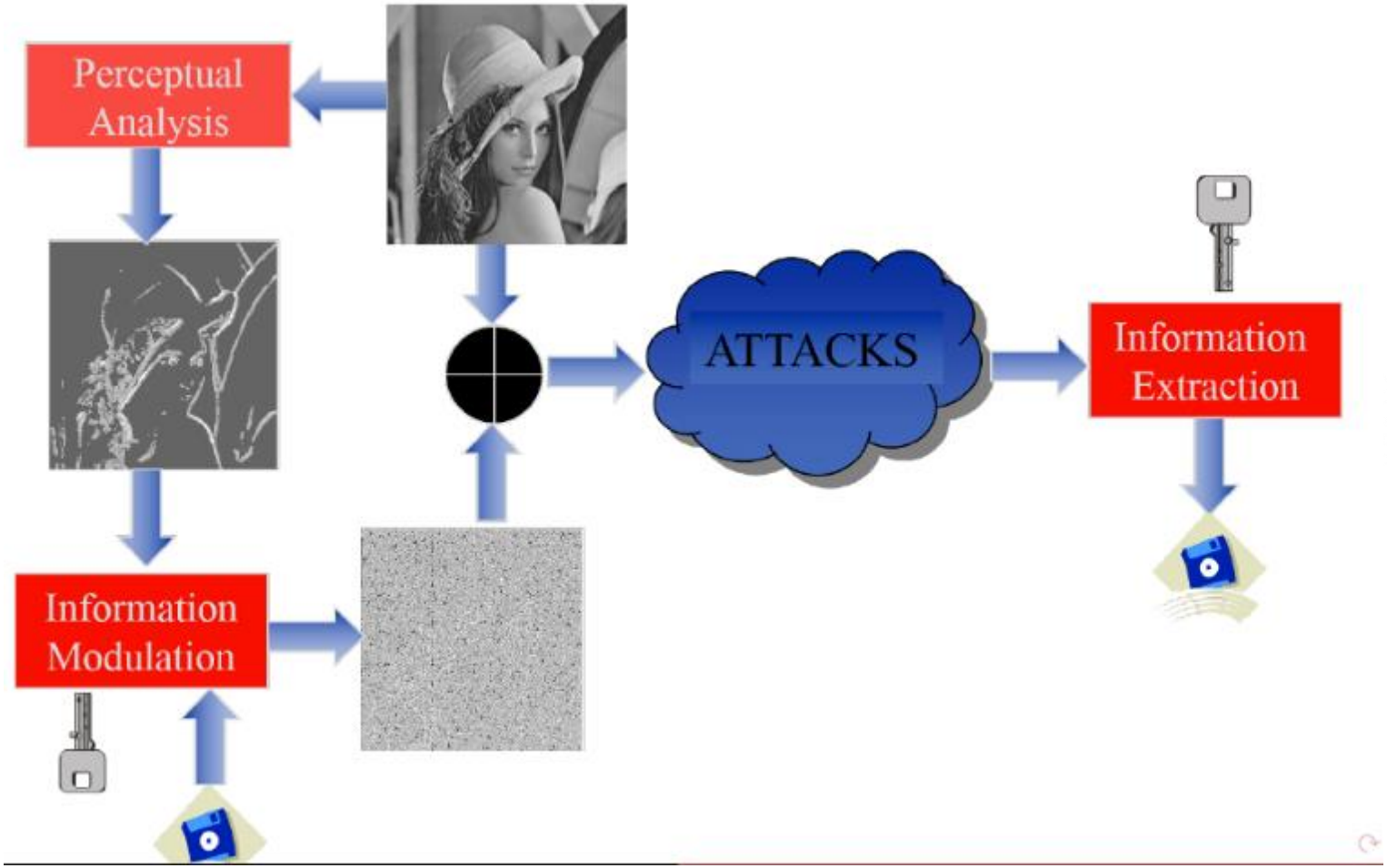
- Host interference present.
- Easy to understand and analyze.

Quantization-based (Chen and Wornell in 1999): the host is quantized with an information-dependent quantizer.

- No host-interference present.
- A bit more difficult to understand and much more difficult to analyze.

A combination of both strategies: **Spread-Transform Dither Modulation (STDM)**.

A MODEL FOR DATA HIDING



HIDING ONE BIT …

Most perceptual experiments concern visibility of noise-like signals and limit its local variance.

Add/substract an invisible noise-like watermark to be decoded later

x: luminance of the pixel in the host image
b: symbol to be embedded $b = \{-1, +1\}$

generate $y = x + b \cdot \alpha$

where $w = b \cdot \alpha$ is a watermark. The parameter α defines how much invisible is the watermark.

How to extract the message?

If you know x, you can compute

$$\hat{b} = \mathrm{sgn}(y - x)$$

It can be robust to noise.

BLIND CASE...

If x is not known, it is not possible to recover b

You can think about subtracting the average, i.e.,

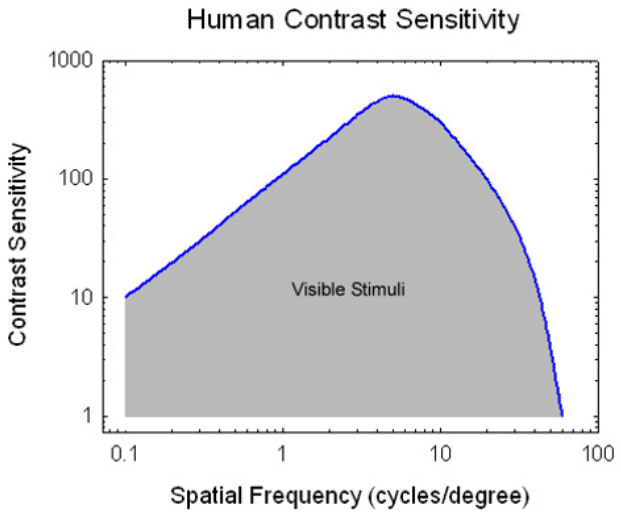
$$\hat{b} = \mathrm{sgn}(y - \mu)$$

Unfortunately, SNR is usually small

$$\sigma(y - \mu) >> \alpha^2$$

You need to spread the spectrum!!!

HUMAN CONTRAST SENSITIVITY



- Contrast sensitivity function (CSF): The human perception system does not respond equally to all spatial frequencies.
- The eye is less sensitive to extremely gradual changes.
- The eye is fairly sensitive to more rapid changes.
- The eye is decreasingly sensitive to yet higher spatial frequencies.

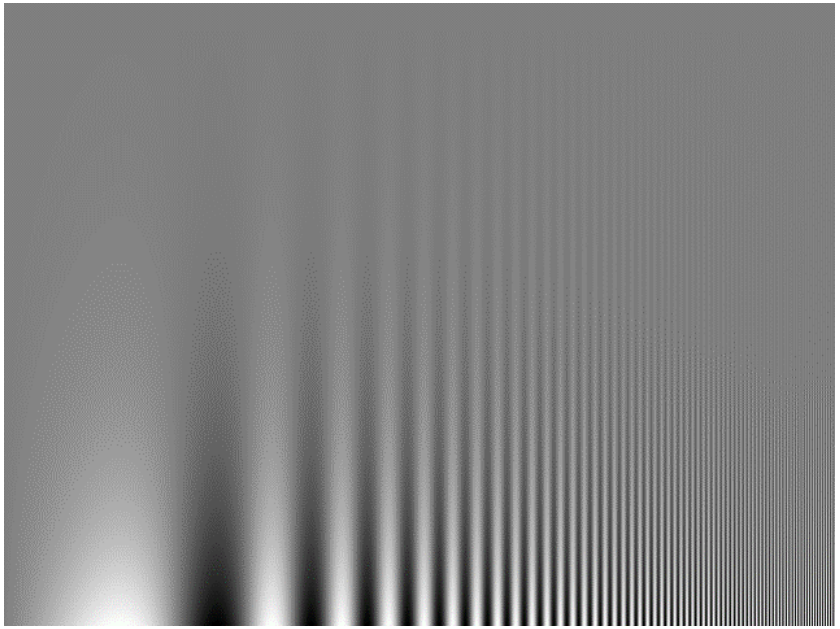
The perceptual mask tells us how much we can modify each pixel so that each modification is equally visible.

Let us focus on the ability of edges to conceal information.

e.g. consider a Gaussian gradient filter.

For each pixel, compute horizontal and vertical gradients h[n1; n2] and v[n1; n2].

$$\alpha[n_1,n_2]=\sqrt{h^2[n_1,n_2]+v^2[n_1,n_2]}$$



EXAMPLES OF PERCEPTUAL MASK

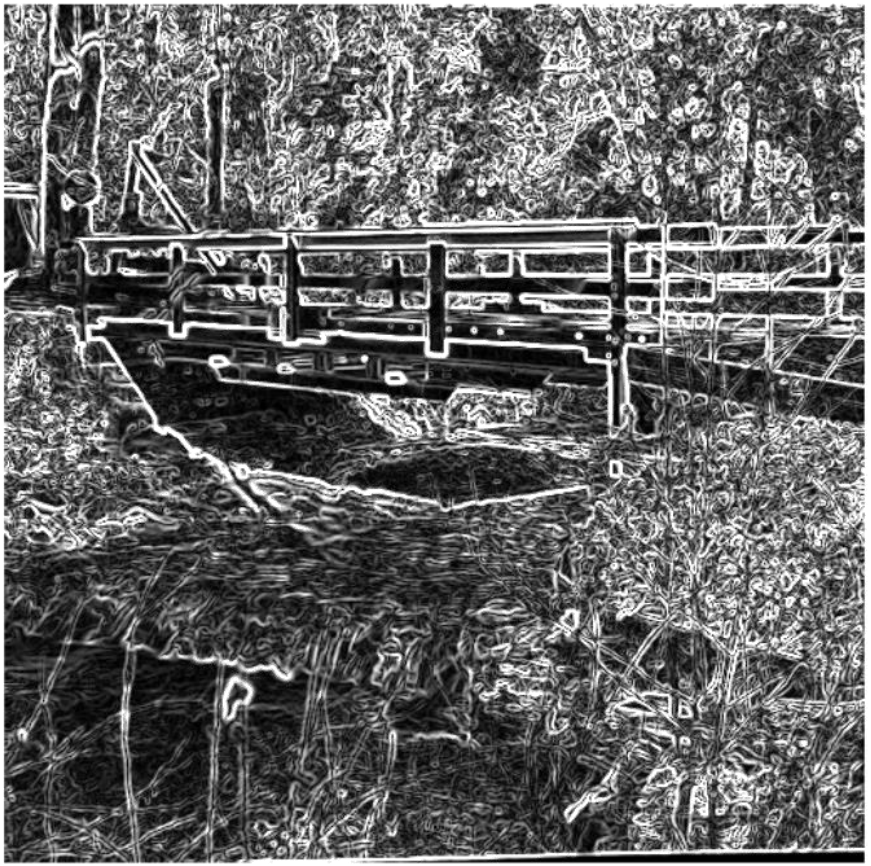




Mask x 10

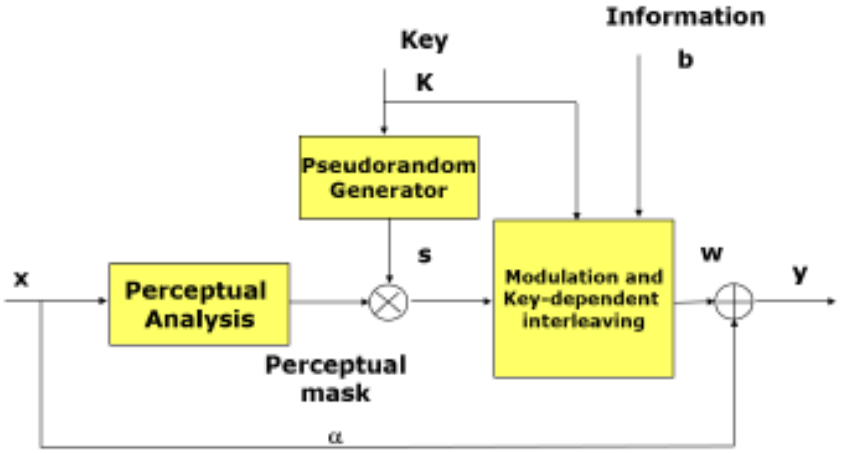
EXAMPLES OF PERCEPTUAL MASK





Mask x 10

MODEL FOR SPREAD-SPECTRUM EMBEDDING



- Assume an image of NxN pixels
- Coordinates $P = \{(n_1, n_2) : n_1, n_2 \in \{1, \ldots, N\}\}$
- Assume a key-dependent tiling

$$T_i : \bigcup_{i=0}^{M-1} T_i = N$$

(each tile T_i is disjoint $|T_i| = \frac{N^2}{M}$

Embedding pulses

$$p_i(n_1, n_2) = \begin{cases} \alpha[n_1, n_2] s[n_1, n_2] & \text{if } (n_1, n_2) \in T_i \\ 0 & \text{otherwise} \end{cases}$$

$\alpha[n_1, n_2]$ Perceptual mask

$s[n_1, n_2]$ Pseud-random sequence with zero-mean and variance 1

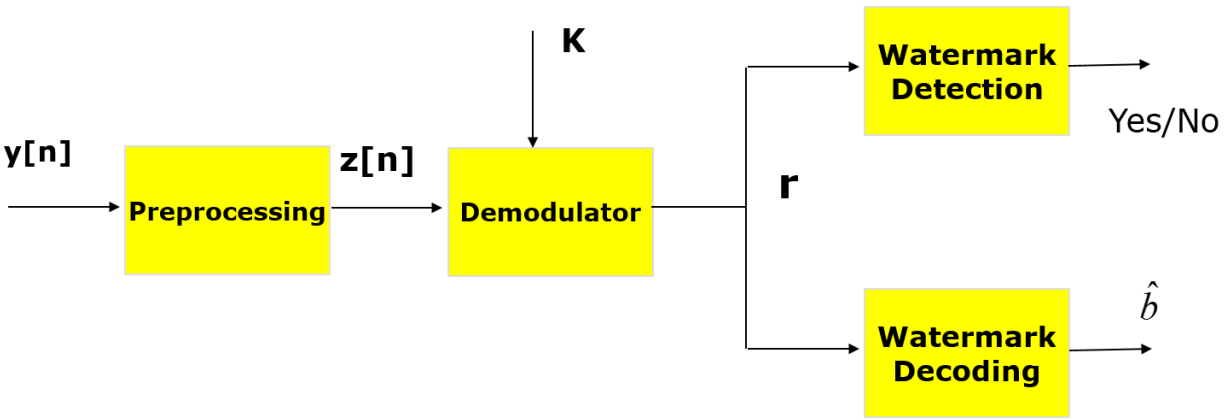
$$w[n_1, n_2] = \sum_{i=0}^{M-1} b_i \, p_i[n_1, n_2]$$ Watermark

Bits of message

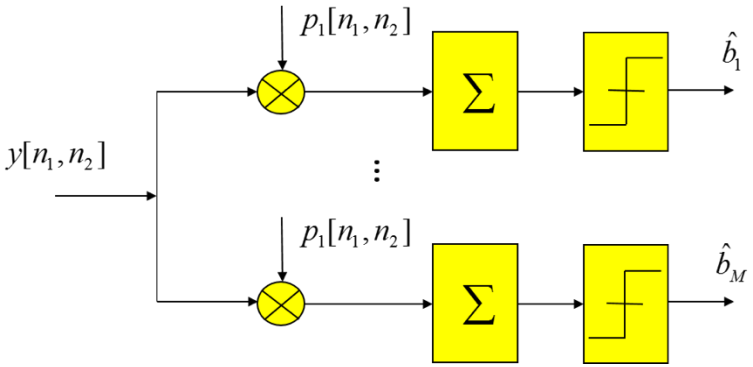
$$y[n_1, n_2] = x[n_1, n_2] + w[n_1, n_2]$$

Final image

DETECTION STRATEGY



Detection is possible without knowing the watermark (of course, the tiling and the pseudorandom sequence must be known).



Let us focus on specific tile (omit i)

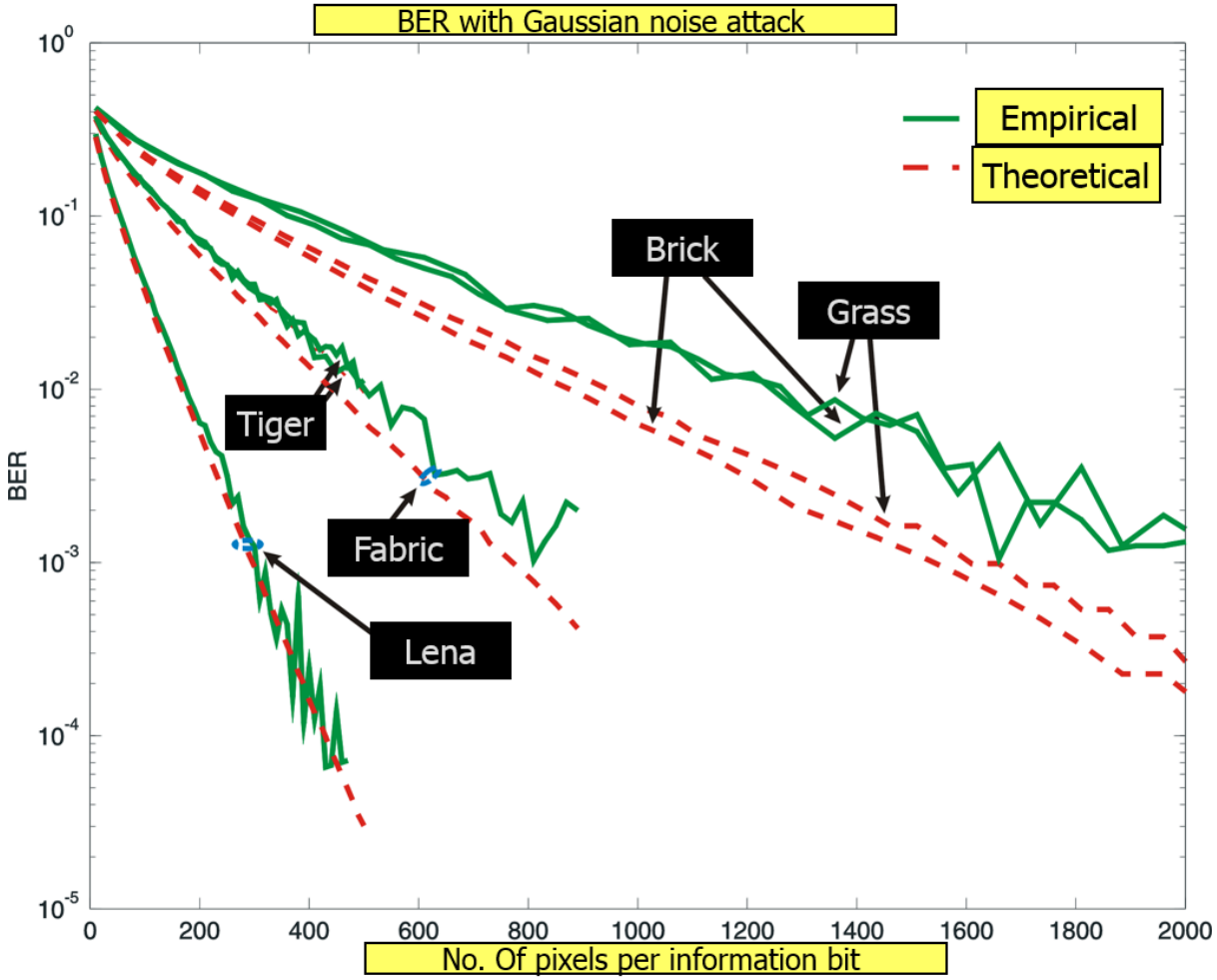
$$y = x + \gamma \, b \, p$$

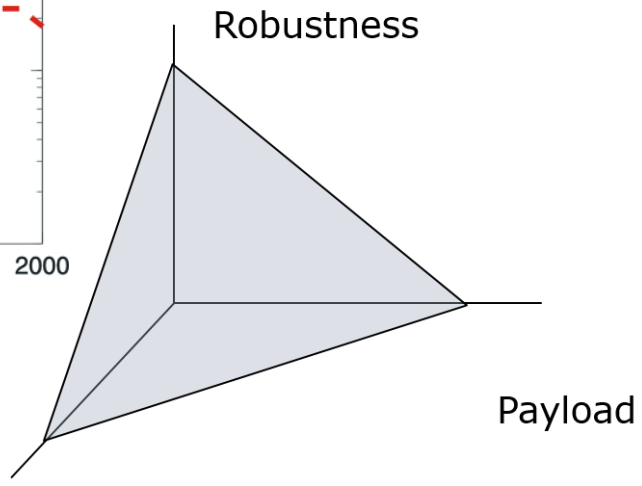
If x i.i.d and Gaussian, optimal decoder is easy to find.

$$\hat{b} = \mathrm{sgn} < y, p >$$

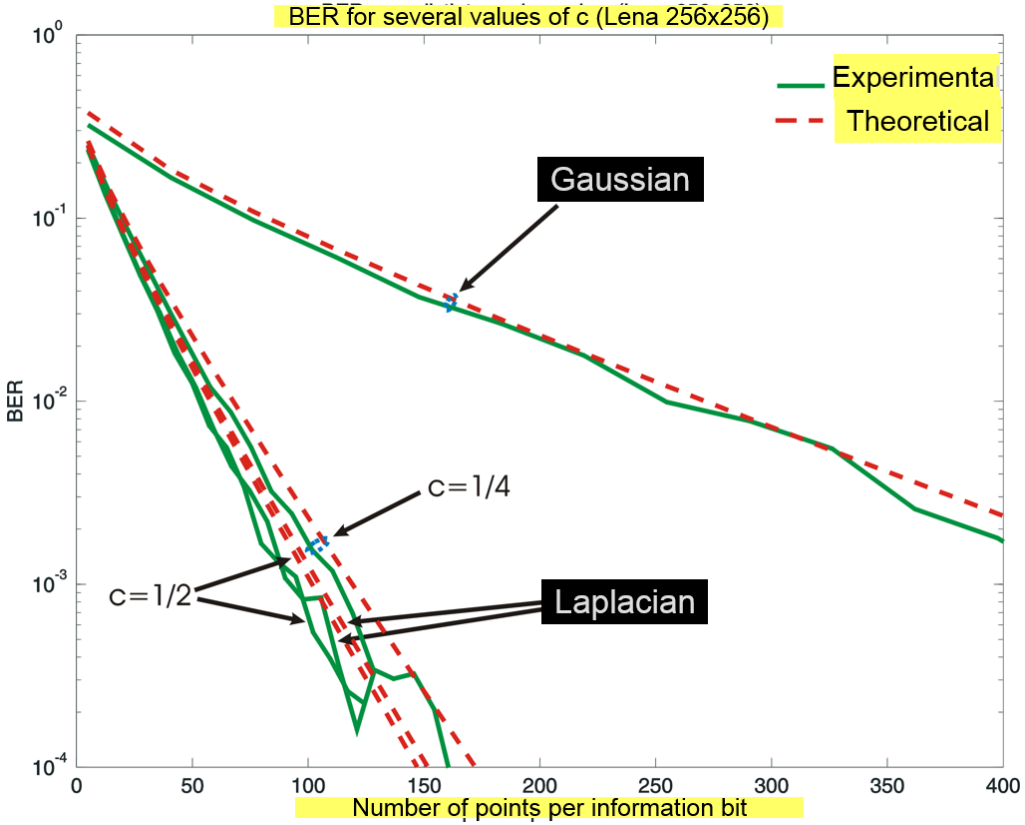
Robustness depends on spreading

PERFORMANCES





PERFORMANCES



ATTACKS

- Additive noise
- Filtering
- Cropping
- Compression
- Rotation and scaling
- Statistical averaging
- Multiple Watermarking

APPLICATIONS

- Video watermarking
- Audio Watermarking
- Software watermarking
- Text watermarking
- Executable watermarking
- Labelling
- Fingerprinting
- Authentication
- Copy and playback control
- Signaling